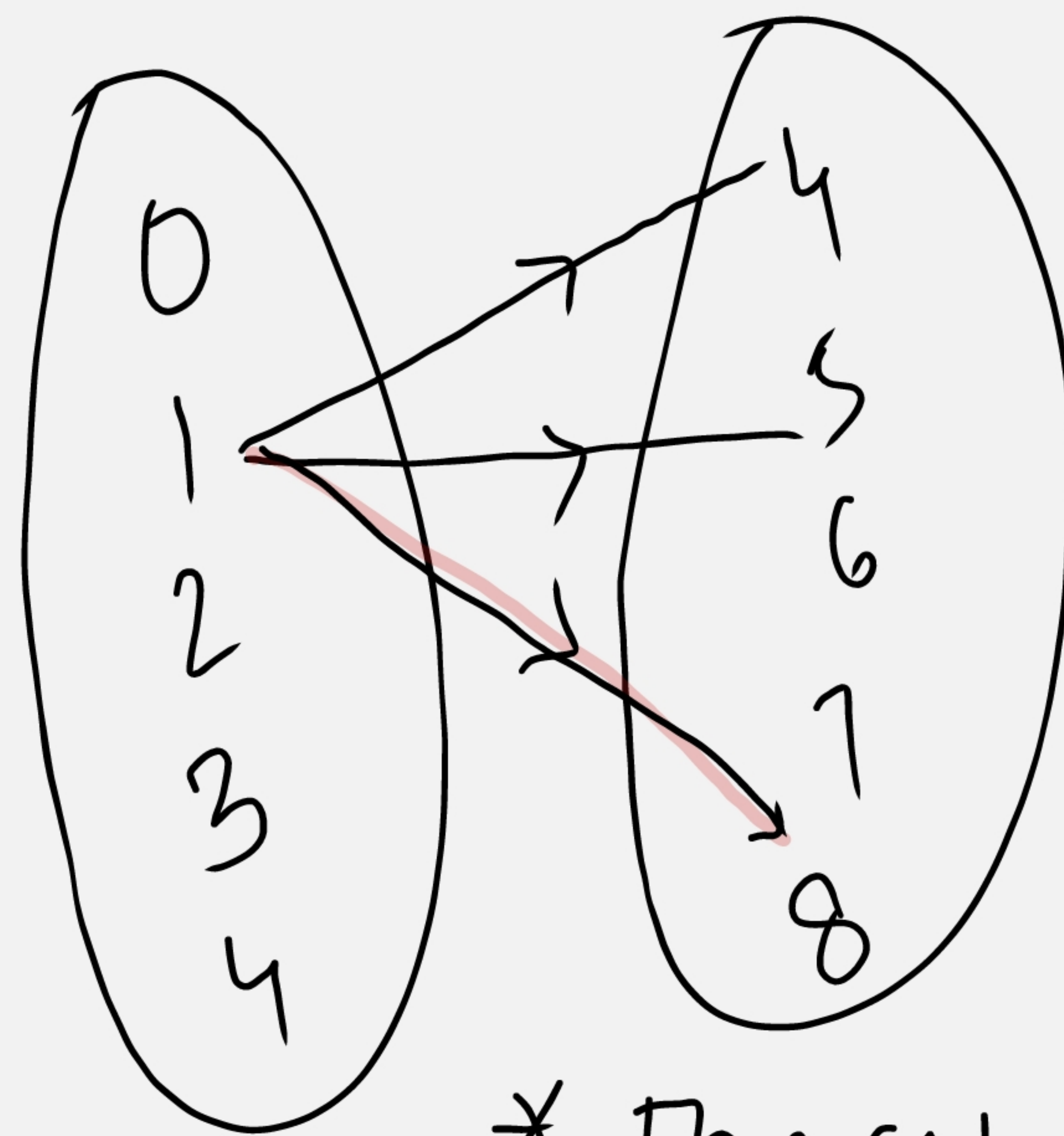


- 1) Equivalence class
- 2) domain & co-domain, Range
- 3) Onto, One-One, Bijective

Equivalence class

Set $A = \{0, 1, 2, 3\}$

Equivalence class of "1" = $\{4, 5, 8\}$



* The set of all elements related to any element $a \in A$ is denoted by $[a]$
 i.e. $[a] = \{x \in A : (x, a) \in R\}$

Hence $[a]$ is called an equivalence class of $a \in A$.

Ex:-

Let R be the equivalence relation in the set $A = \{0, 1, 2, 3, 4, 5\}$ given by
 $R = \{ (\underline{a}, \underline{b}) : 2 \text{ divides } (a-b) \}$. Write the equivalence class of $[0]$.

Note:-

" m " divides by " n "

$\approx$ " n " is divisible by " m "

$\equiv \frac{n}{m} = \text{Integer}$

$$\frac{a-b}{2} = \text{Integer}$$

$$(0, 0) \text{ or } (0, 0)$$

$$(0, 2) \text{ or } (2, 0)$$

$$(0, 4) \text{ or } (4, 0)$$

$$\text{Equivalence class of } [0] = \{0, 2, 4\} \checkmark$$

Question:-
Assertion(A): In a set $A = \{1, 2, 3\}$ a relation R defined as $R = \{(1, 1), (2, 2)\}$ is reflexive.

Reason(R): A relation R is reflexive in set A if $(a, a) \in R$ for an $a \in A$ ✓

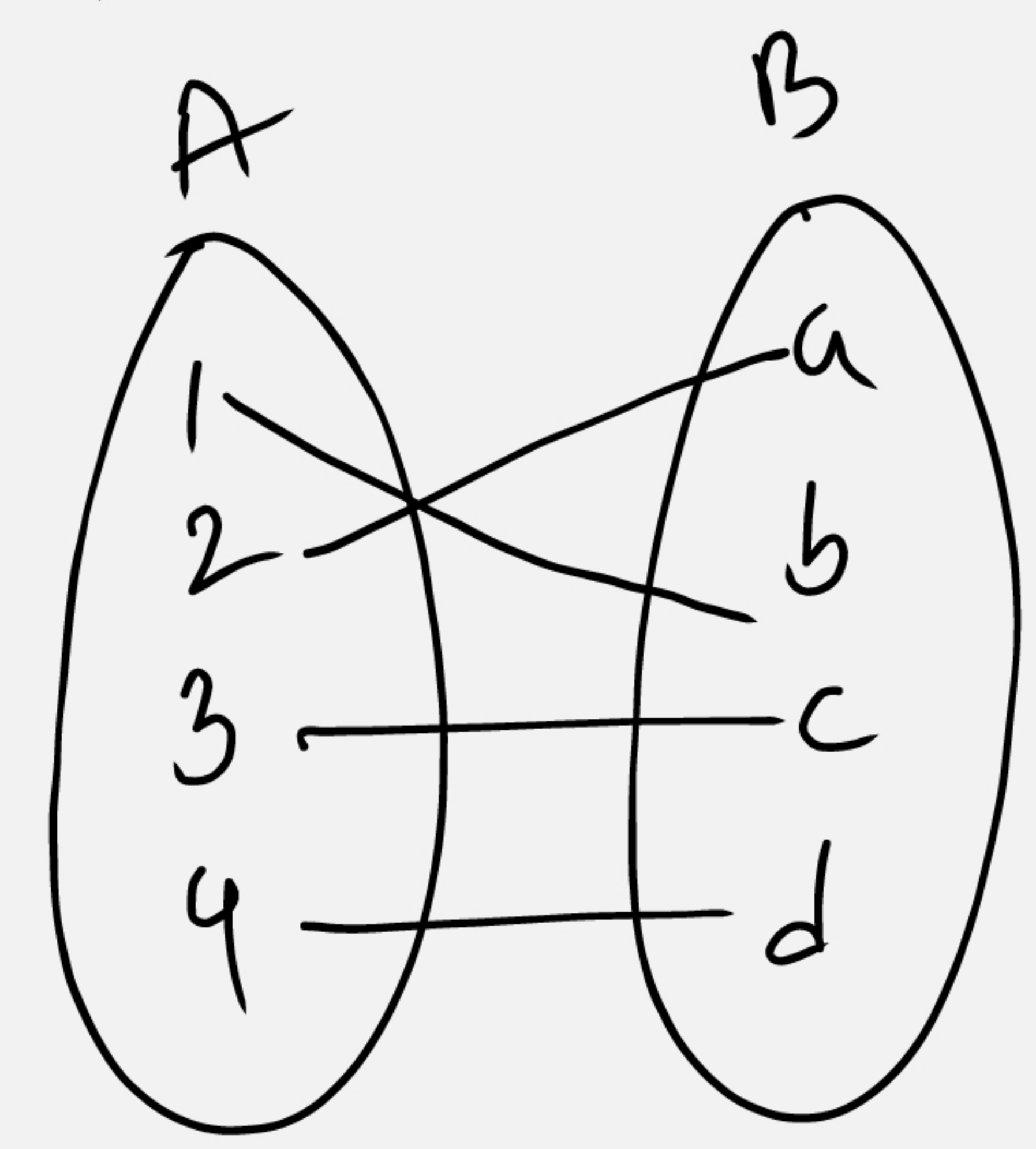
- (A) Both A and R are true and R is correct explanation of A ✓
(B) Both A and R are true but R is not the correct explanation of A .
(C) A is true but R is false
(D) A is false but R is true.

Special Type of function

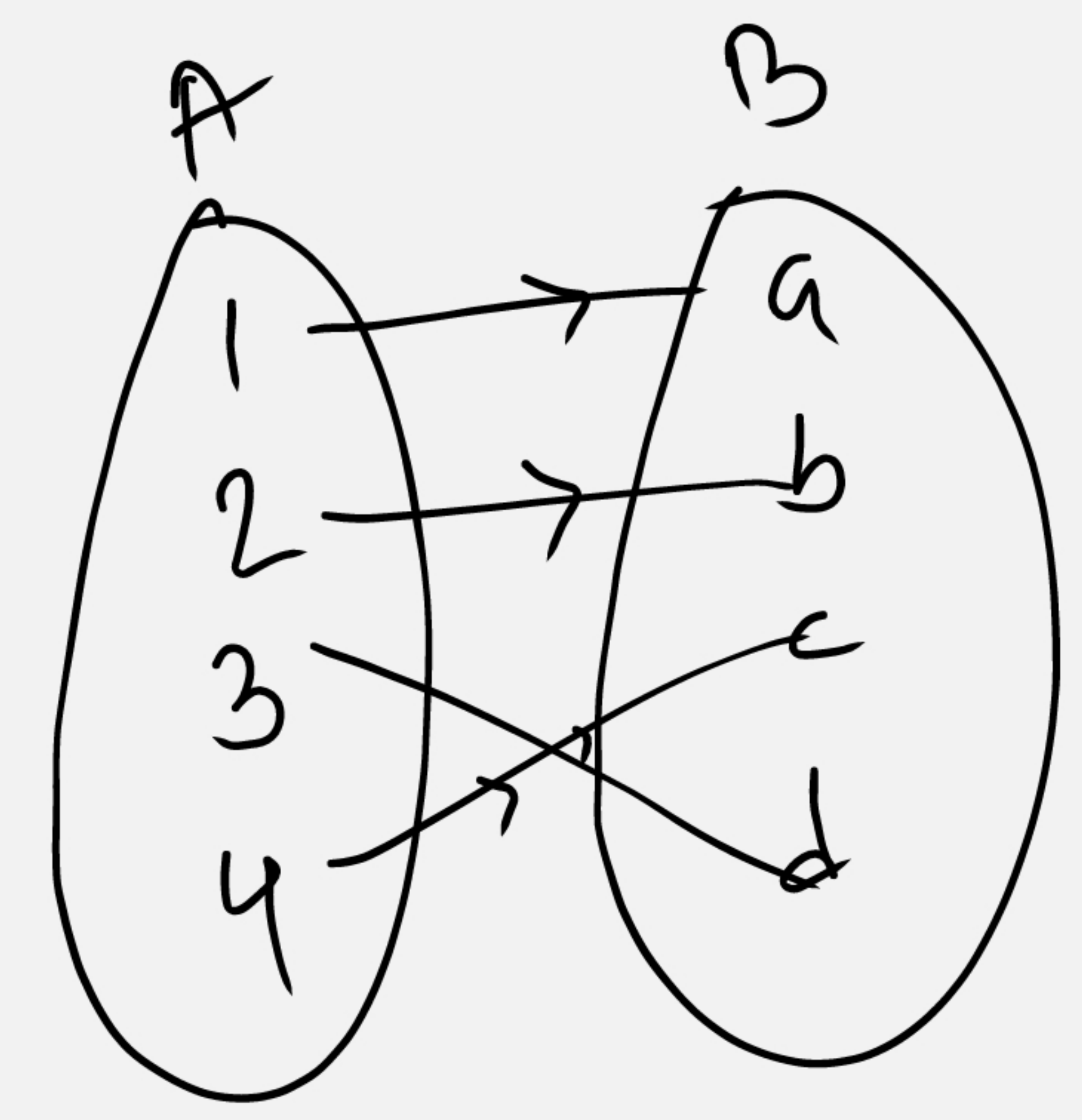
One-One Function :-

Every element of A has distinct image in B .

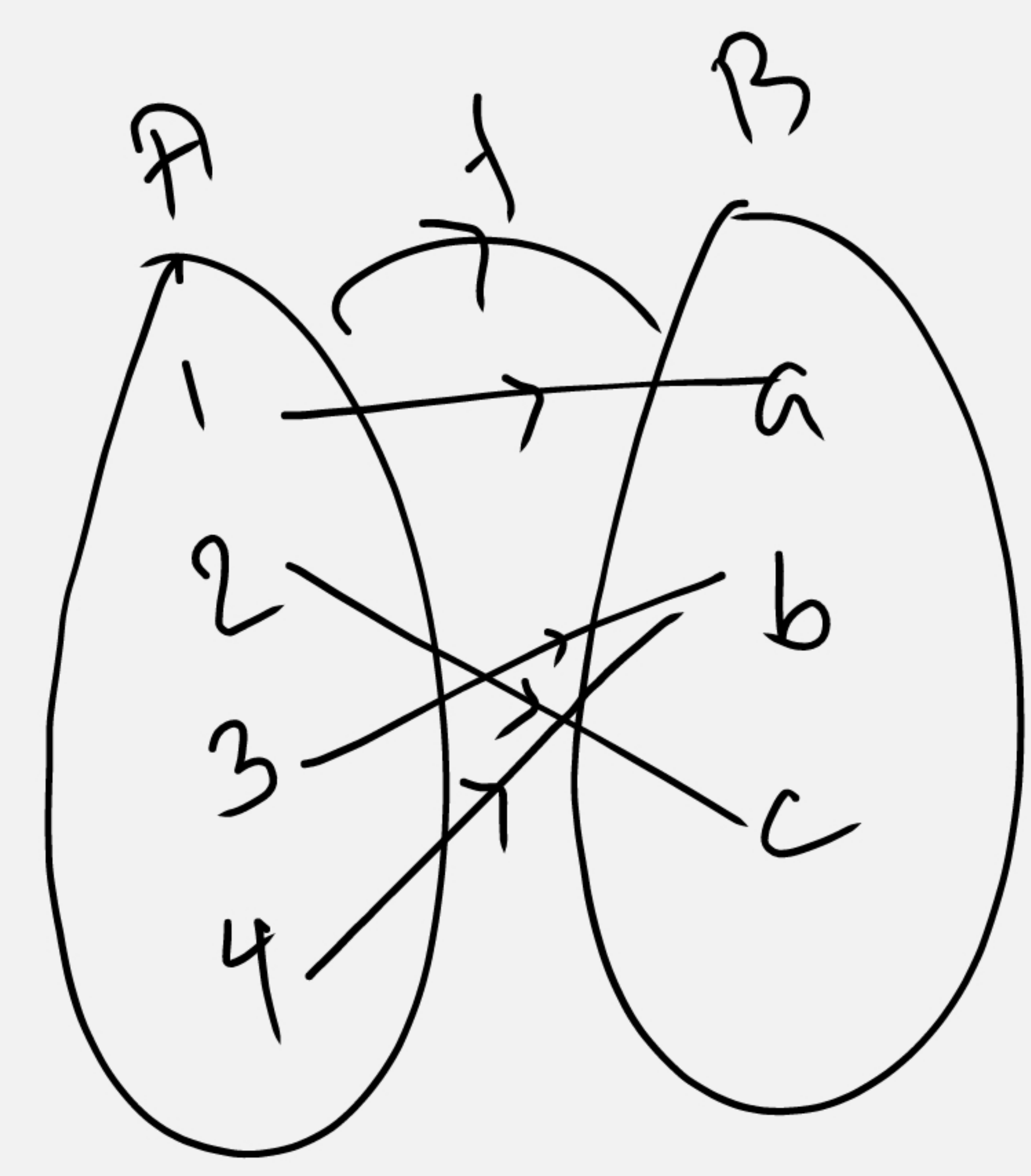
injective



one-one



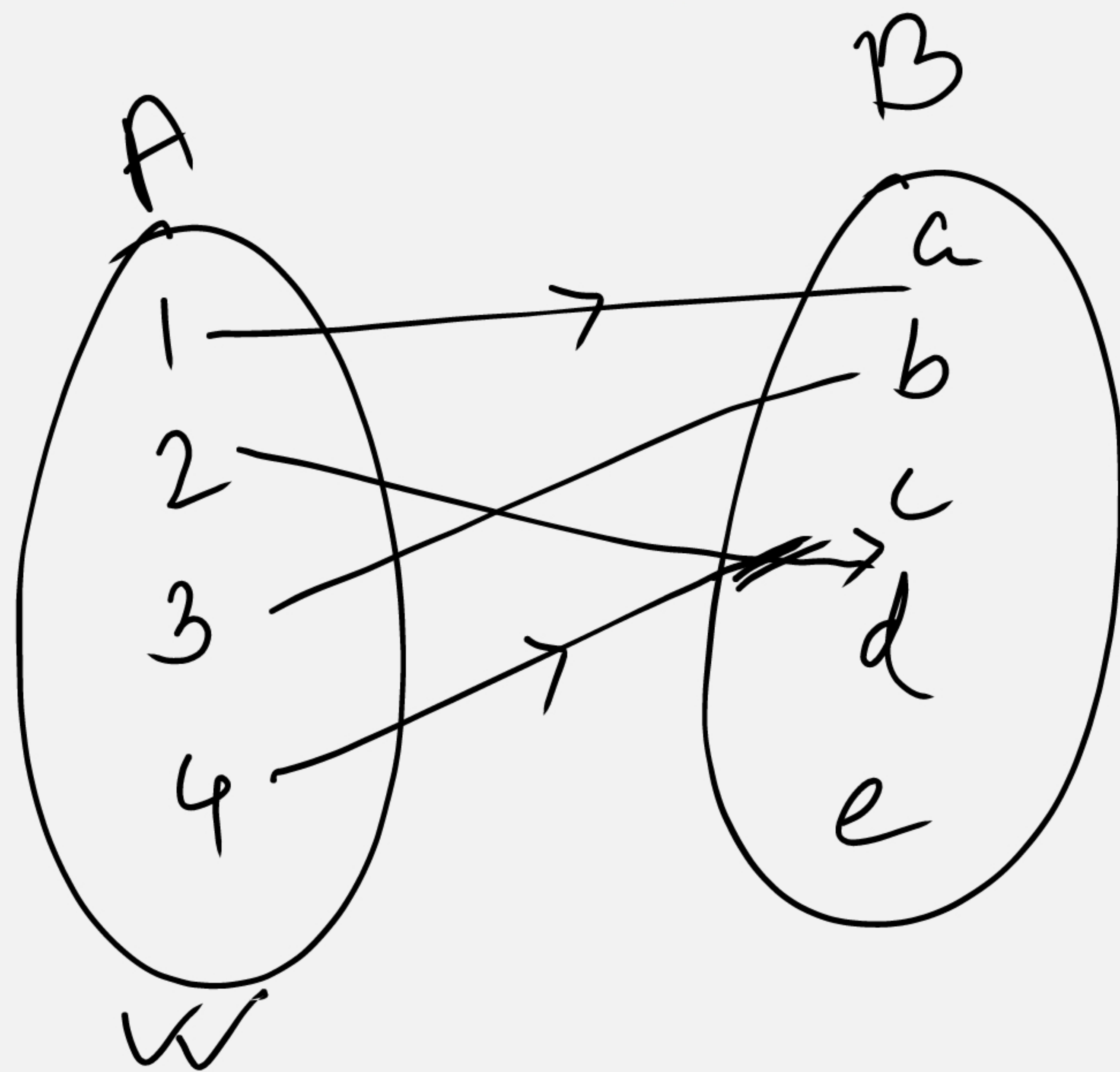
one-one



not one-one

Many to one Function

Two or more elements of A has same image in B.



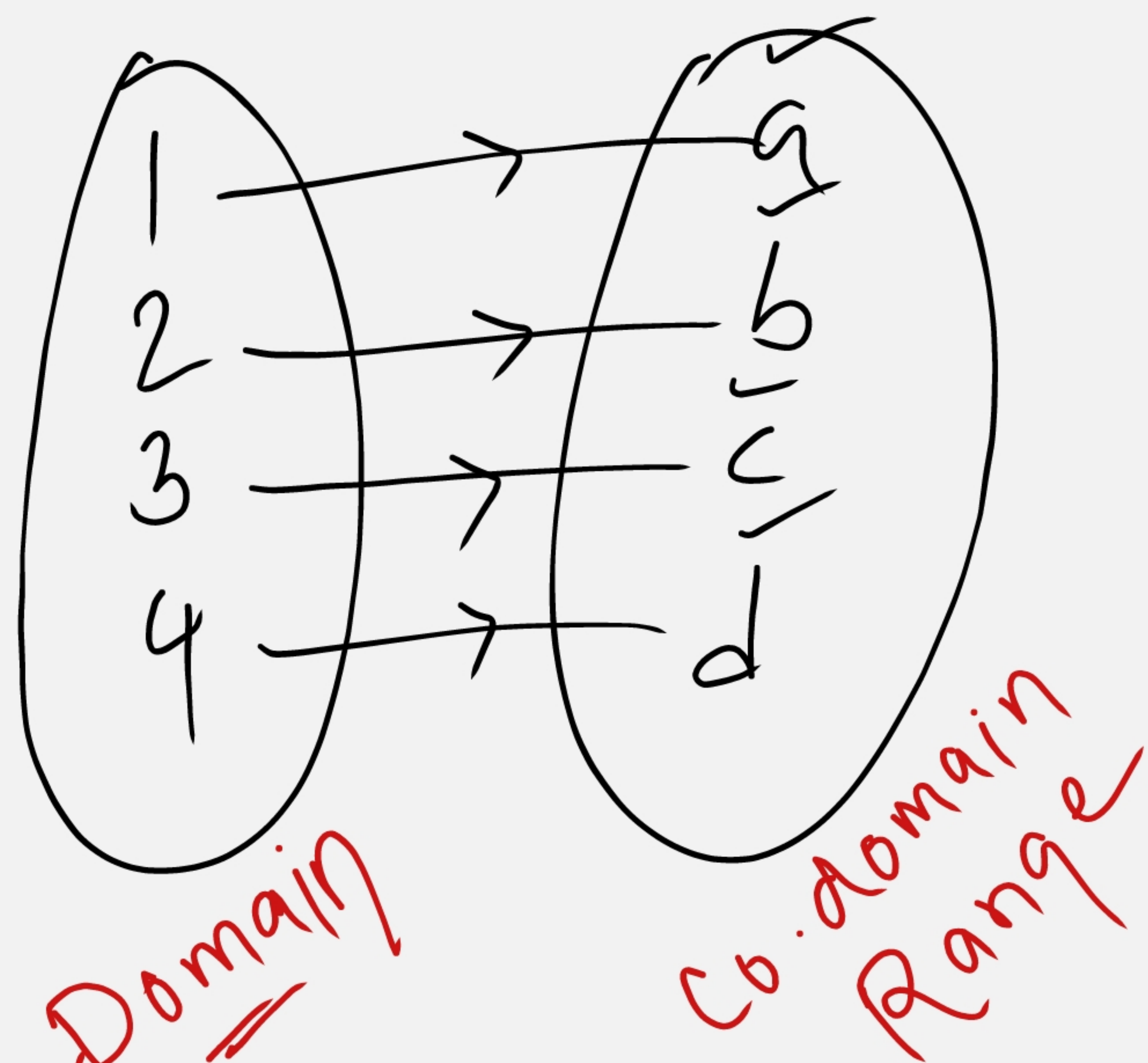
one-one

converge

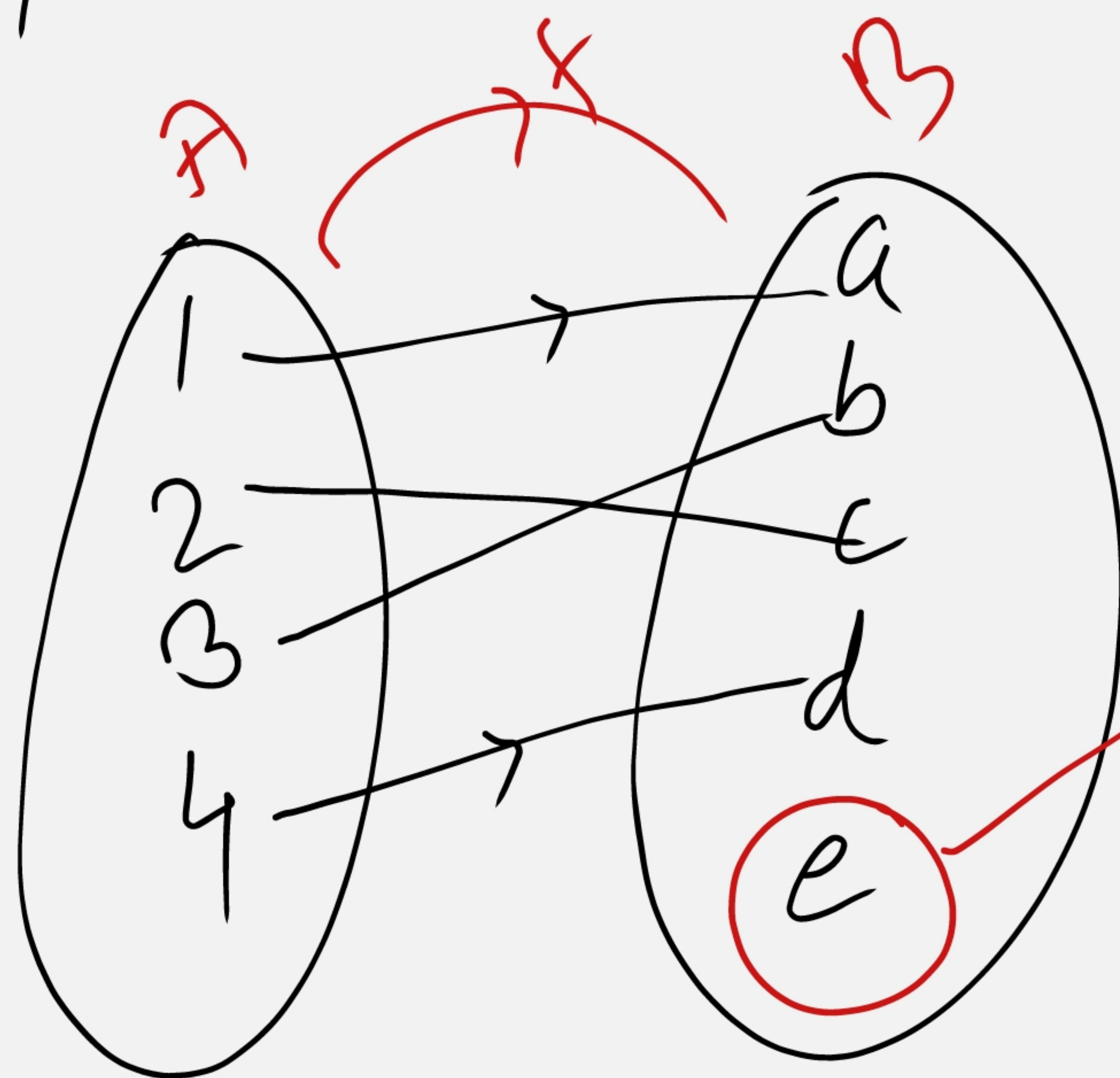
one to many / many to one //

Onto-Function
(Surjective) $\rightarrow B(x)$ single

Every element in B has atleast one preimage in A .



Range $\neq$ co-domain $\Rightarrow$ Into ✓

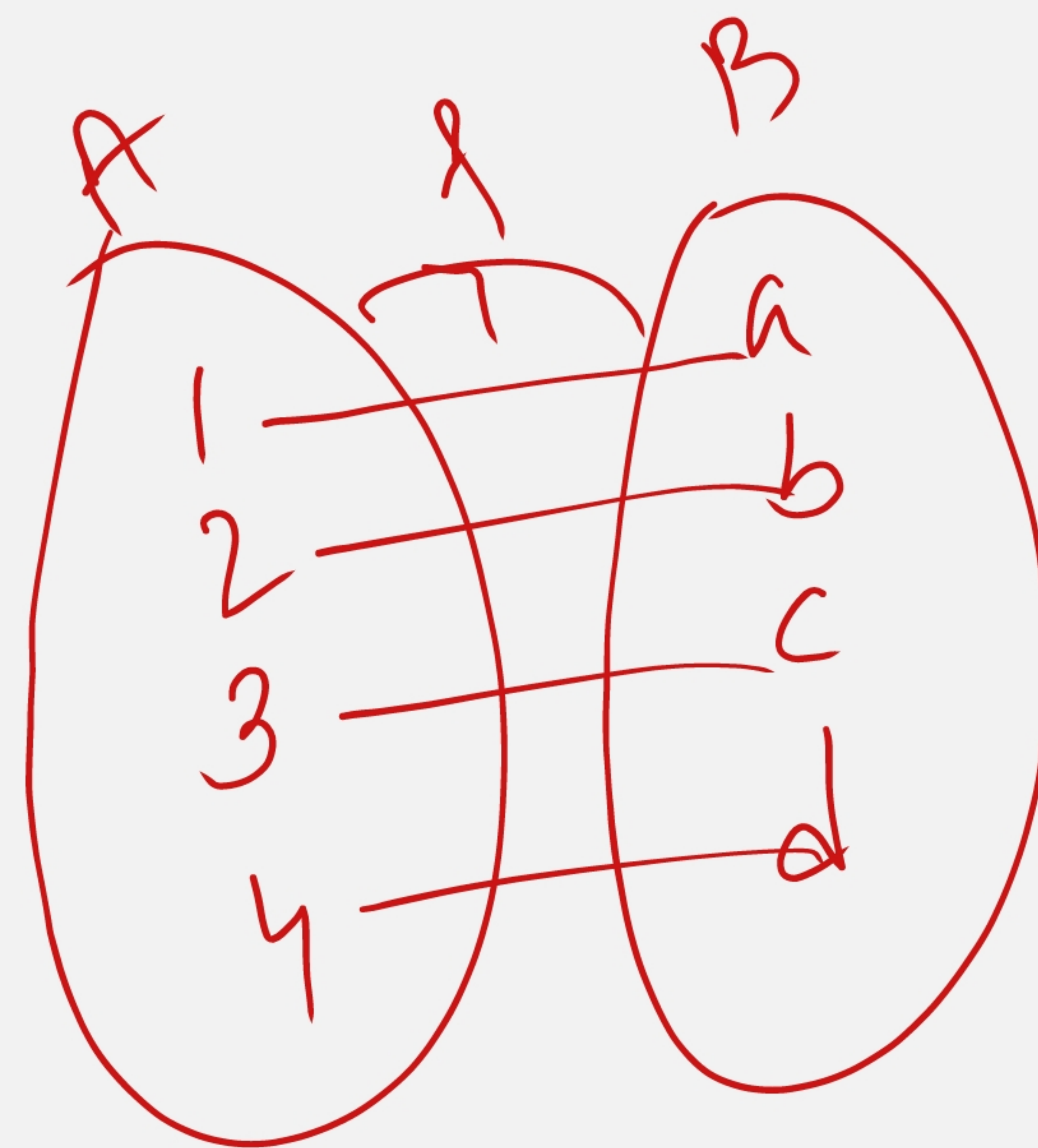
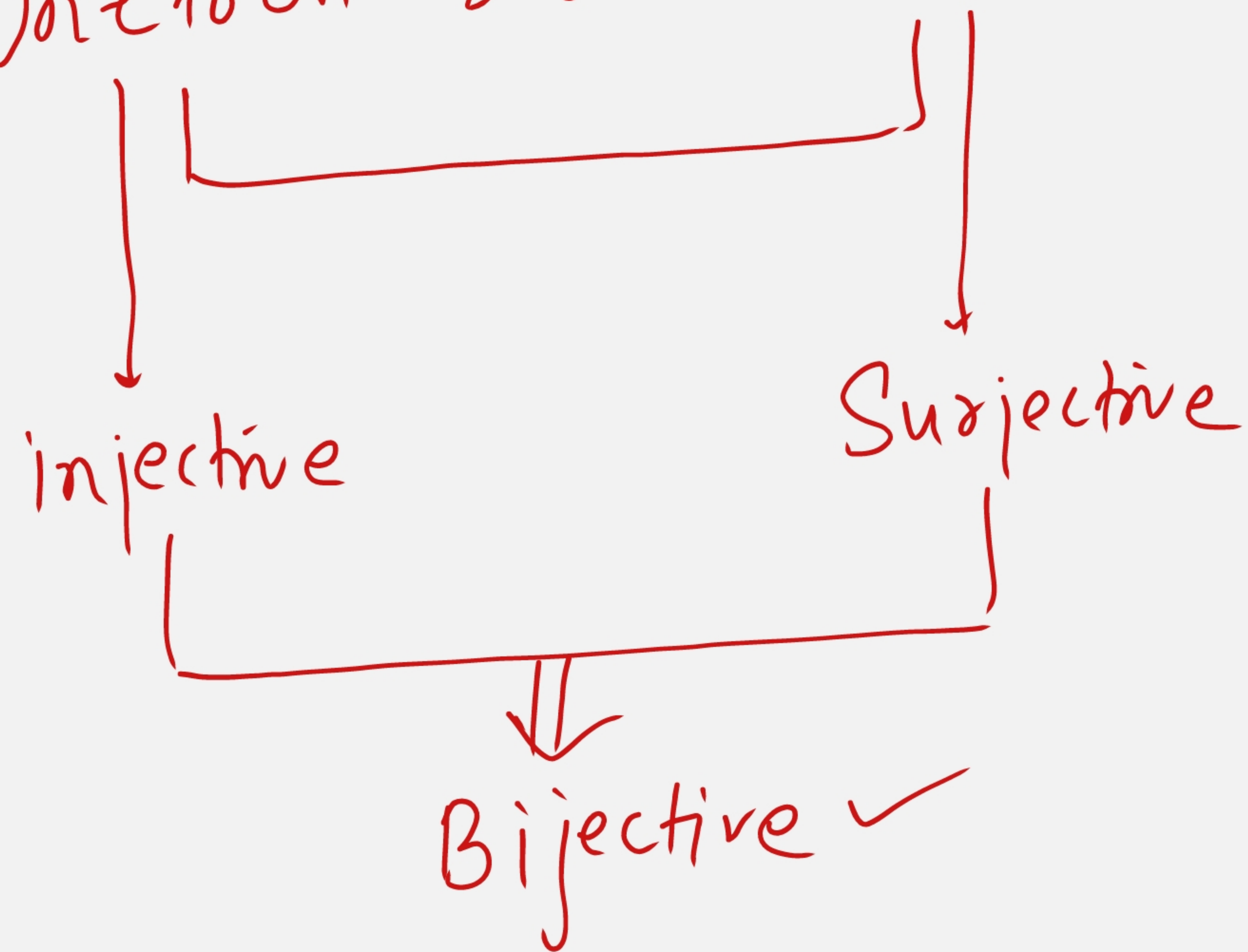


Range = $\{a, b, c, d\}$
co-domain = $\{a, b, c, d, e\}$ $\Rightarrow$ Surjective ✓

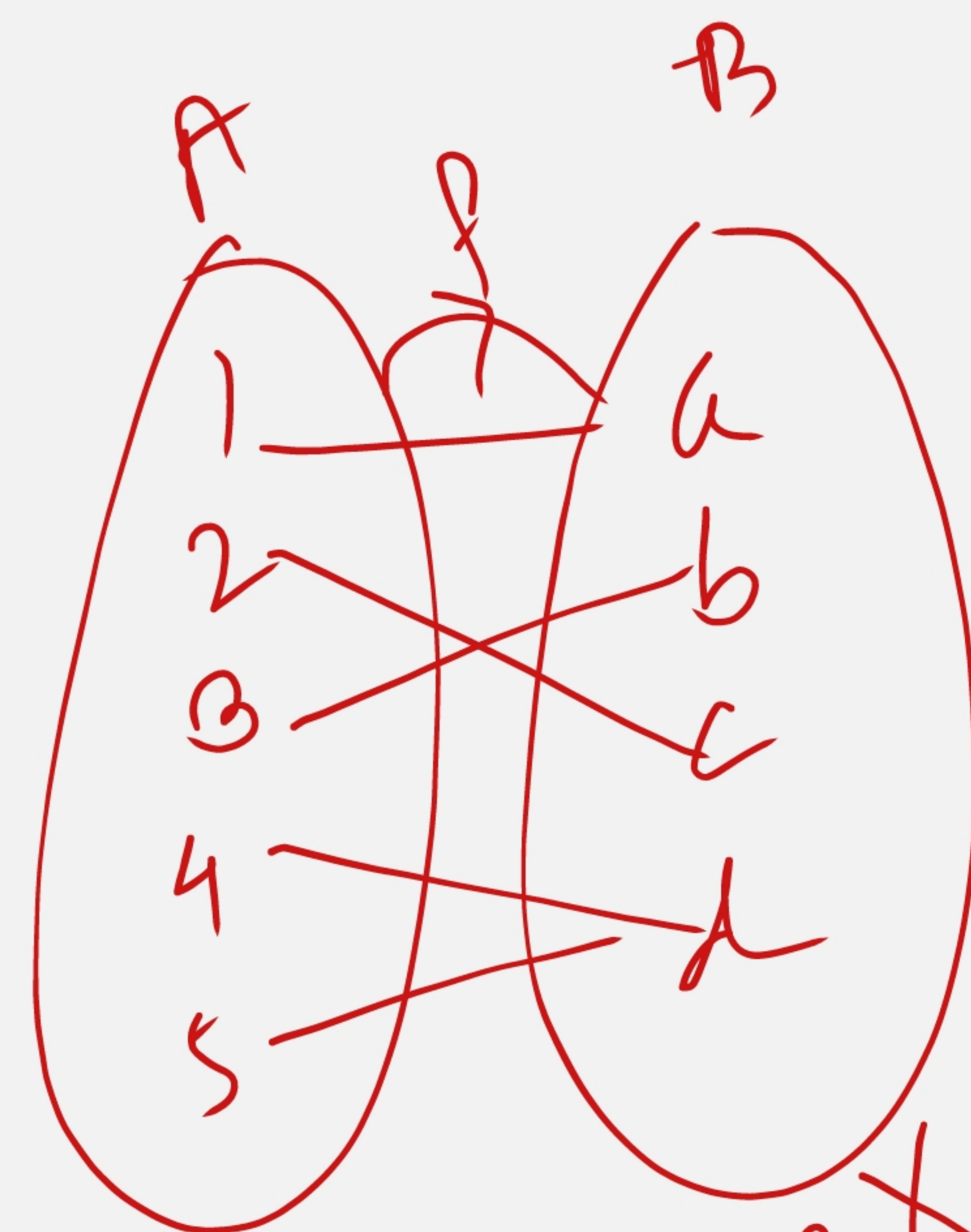
Note :

single
Not onto
Surjective X
Into

One-to-one & Onto Function



Bijective

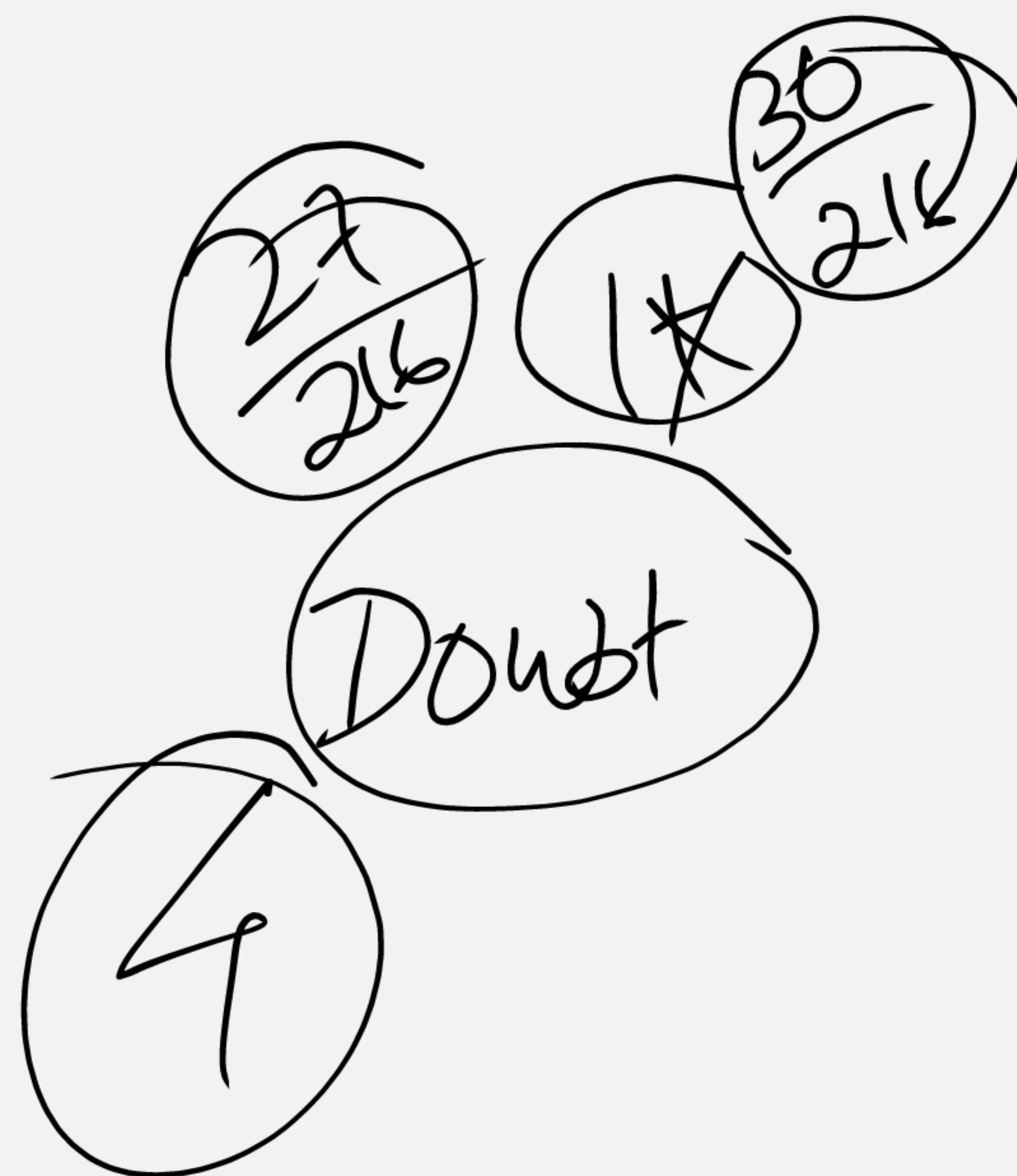


one-one /
Not Bijective ✓

The no. of functions defined from $\{1, 2, 3, 4, 5\} \rightarrow \{a, b\}$ which are one-one?

- a) 5
- b) 3
- c) 2
- ~~d) 0~~

Why?



Friday

Topic :-

- 1) Probability Basics
- 2) Independent & Dependent
- 3) Mutually Exclusive & Inclusive
- 4) $10K1 = 10$

